K. V. Jobin, Ph.D.

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PROFESSIONAL SUMMARY

PhD researcher and Senior AI/ML Engineer with expertise in vision-language architectures, multimodal document comprehension, and agentic systems. Proven track record of publishing award-winning research at top-tier conferences (WACV, CVIP, ICDAR) and translating foundational research into enterprise-grade AI solutions. Specialized in deep learning, transformer-based search, semantic layout analysis, and autonomous agent orchestration.

RESEARCH INTERESTS

- **Multimodal & Vision-Language AI:** Document image analysis, semantic layout understanding, visual grounding, and cross-modal Retrieval-Augmented Generation (RAG).
- **Reasoning & Agentic Systems:** Multi-agent orchestration, tool integration, stateful workflows, and autonomous decision-making systems.
- **Foundation Models & Efficient AI:** Vision Transformers (ViT), fine-tuning multimodal foundation models, and scalable enterprise deployment architectures.

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Bourntec Solutions **2024 – Present**

- Architecting agentic AI frameworks and multimodal integration pipelines for real-world enterprise applications.
- Designing scalable LLM/agentic workflows, stateful multi-agent systems, and local/enterprise AI deployments prioritizing performance and data security.

Graduate Researcher | IIIT-Hyderabad **2015 – 2025**

- Conducted cutting-edge research in Document Image Analysis, Computer Vision, and Machine Learning.
- Developed Semantic Labels-Aware Transformer models for multimodal search over large slide collections (WACV Paper Award Finalist).
- Created benchmark datasets (DocFigure) and deep learning models for scientific figure classification and multi-modular document analysis.

Trainee Engineer | IIIT-Hyderabad **2013 – 2015**

- Contributed to the MCIT (Govt. of India) project developing multilingual OCR systems for Indian languages.
- Enhanced document layout segmentation algorithms for complex, real-world document structures.

EDUCATION

Ph.D. in ECE (Computer Vision, ML & AI) | IIIT-Hyderabad **2013 – 2023**

- *Thesis: Document Image Layout Segmentation and Applications*

M.Tech. in Signal Processing | CET Trivandrum **2011 – 2013**

- *Thesis: Automatic Number Plate Detection and Extraction*

B.Tech. in Electronics and Communication | CEA Adoor **2006 – 2010**

SELECTED RESEARCH PUBLICATIONS

Conference Proceedings

- **K. V. Jobin**, A. Mishra, and C. V. Jawahar. "Semantic labels-aware transformer model for searching over a large collection of lecture-slides." IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2024. **(Paper Award Finalist)**
- **K. V. Jobin**, A. Mondal, and C. V. Jawahar. "Classroom slide narration system." CVIP, 2021. **(IAPR Best Paper Award)**
- **K. V. Jobin**, A. Mondal, and C. V. Jawahar. "DocFigure: A dataset for scientific document figure classification." ICDARW, 2019.
- **K. V. Jobin** and C. V. Jawahar. "Document image segmentation using deep features." NCVPRIPG, 2017.
- **K. V. Jobin**, C. V. Jiji, and P. R. Anurenjan. "Automatic number plate recognition system using modified stroke width transform." NCVPRIPG, 2013.

Journal Articles

- **K. V. Jobin**, A. Mondal, and C. V. Jawahar. "Document image analysis using deep multi-modular features." *SN Computer Science*, vol. 4, no. 1, p. 5, 2022.

HONORS & AWARDS

- **Paper Award Finalist –** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
- **IAPR Best Paper Award –** 6th IAPR International Conference on Computer Vision & Image Processing (CVIP), 2021.
- **Visvesvaraya PhD Fellowship –** IIIT-Hyderabad, 2015.
- **Runner-Up –** Microsoft Azure ML Hackathon, IIIT-Hyderabad, 2011.
- **GATE Scholarship –** CET, Trivandrum, 2011.

TECHNICAL SKILLS

- **Core AI/ML Areas:** Multimodal Learning, Vision Transformers, Foundation Models, Document Image Analysis, Agentic Systems, Cross-Modal RAG.
- **Frameworks & Tools:** PyTorch, Keras, OpenCV, Scikit-learn, LangChain/LangGraph, Hugging Face Transformers.
- **Languages & Infra:** Python, MATLAB, C++, Java, SQL, Linux, Docker, LaTeX Typesetting & Publishing.